

A Wearable Haptic Sleeve Interface for Spaceflight Robotic Arm Operations

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ABSTRACT

During spacecraft robotic arm operations, haptic feedback has the potential to display rich streams of relevant data to the user, such as trajectory information and velocity. One promising strategy to encode such information is through tactile stroke illusions, which create the sensation of vibration moving across the skin. This study highlights the design and feasibility testing of a haptic wearable interface worn on the user's forearm in a sleeve form factor that elicits tactile stroke illusions, using an array of vibration actuators in a timed sequence. By manipulating the temporal aspects and developing two rendering algorithms to convey end-effector movement information, we demonstrate the feasibility of our haptic sleeve wearable by human testing to measure how accurately participants perceive the conveyed information (trajectory and velocity).

CCS CONCEPTS

• **Human-centered computing** → **Usability testing**.

KEYWORDS

Haptic Feedback, Space-Robotic Operations, Tactile Stroke Illusions, Wearable Device

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1 INTRODUCTION

Robotic arm operations in spaceflight or on the Lunar surface impose significant operational demands on space crew members. The process of dynamically integrating a multitude of sources providing the visual information required to safely maneuver the robotic arm to its next target configuration is particularly challenging for crew members. Beyond the real chance of overloading the operator's visual sensory channel, challenges can be compounded in the long-term isolated environments, given the known sensorimotor neurological decline that crew members may experience with duration in isolation and microgravity environments [32],[26]. These neurological changes can increase overall risk and potentially lead to suboptimal performance when operating the robotic arm.

Haptics, which refers to both real and simulated touch interactions, can provide supplementary sources of information to visual sensory feedback. This feedback can help manage the cognitive load associated with robotic operations, engage brain areas known to be affected during long-duration space missions [26],[28], and potentially improve operator's performance by relieving the excessive dependency on the highly taxed visual sensory channels [1],[4],[7].

Recent advancements in wearable technology have made addressing these demands more feasible[37, 38]. However, wearable haptic interfaces that hold tightly to the user's skin hold significant promise in microgravity environments, where contact with the skin is minimal as even clothing floats around the user's body[15, 16, 36]. Vibrotactile feedback has been widely used in many relevant wearable applications for notifications including operational cueing for teleoperated robot control in terrestrial applications [17], in-flight cueing for aircraft manual guidance and control [29], providing information for immersive experiences in virtual reality [39], and enhancing haptic experiences in storytelling [22].

In this study, we present the design and feasibility testing of a wearable haptic interface in a sleeve form factor tailored for spaceflight robotic arm operations. Our system can be worn on the operator's upper limb and encode information by evoking haptic stroke illusions using an array of vibrotactors. We performed human testing to demonstrate the system's feasibility in providing movement

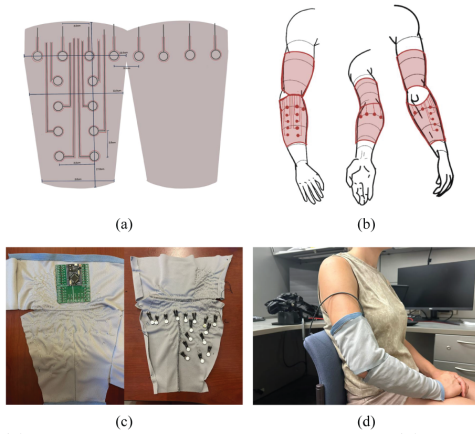


Figure 1: (a) Schematic orientation of vibrotactors, (b) wearable haptic sleeve, and (c) e-textile signal transmission and vibrotactor connections (d) a haptic sleeve on the forearm

information to robotic arm operators, including three-dimensional trajectory and velocity feedback.

2 BACKGROUND

Tactile stroke illusion describes the sensation of moving tactile sensation achieved through the sequential activation of vibration patterns [11]. This technique strategically manipulates the physical characteristics of vibrations to convey a wealth of information. It has shown wide applicability across a variety of fields including navigational guidance and virtual reality applications [24],[35]. This phenomenon is grounded in two separate yet related psychophysical models of touch illusions: (1) apparent tactile motion (ATM), and (2) phantom sensation (PS) illusions. ATM occurs when the activation timing of two (or more) closely positioned vibrotactors on the skin overlaps, creating the sensation of a single wave of vibration passing between them [5],[30]. PS describes the experience of a single “virtual” vibration source perceived between two physical vibrotactors, and the location varying depending on their relative vibration amplitudes [33],[14]. Collectively, these illusions have demonstrated a promising path to generating rich tactile experiences in a compact and efficient manner.

The upper limb is one of the body parts that can efficiently receive haptic feedback. Firstly, the skin on the upper limbs possesses some of the highest densities of mechanoreceptors (biological tactile sensors) and one of the greatest tactile acuities found on the body [34]. Furthermore, the brain is finely tuned to receive tactile information here as these areas have the largest representation in the primary somatosensory cortex [2]. Many studies have explored the impact of vibration frequency on the recognition and perception of cutaneous vibration stimuli. Most studies have highlighted that vibration frequencies between 30 - 350Hz can be readily detected, and these values align with the excitation frequency range of Meissner and Pacinian corpuscles, the skin-based mechanoreceptors responsible for sensing vibration [27],[3]. Optimal sensitivity has been reported to occur at vibration frequencies between 150 and 300Hz across various body sites [23]. While variations in the amplitude of vibration can also be harnessed for conveying a rich stream of information, research focusing solely on the influence

of amplitude remains relatively unexplored. This is primarily due to the common use of eccentric rotating mass vibrotactors in the literature, where the physics of these systems means that vibration frequency and amplitude cannot be separated and controlled independently of one another.

Finally, the arrangement of vibrotactors is also an important factor to be considered for conveying reliable and latent information. As an ability to discriminate individually between two vibration sources is dependent on the dermatomal representation of the upper arm, a two-point discrimination distance can drive the selection of spacing between multiple vibrotactors. Notably, the human capacity to discriminate spatial gaps between two vibration actuators applied to the upper limb falls within the range of 3cm to 5cm [19],[8],[31]. Furthermore, the maximum threshold for each part of the upper limb (forearm: 6cm - 7cm, upper arm: around 7cm) has been established for inducing phantom sensation. It is also important to consider the inherent diversity in anthropometric measurements among individuals’ upper and forearms for better perception of vibration [10].

3 DESIGN STATEMENT

3.1 Prototype Development

The wearable haptic sleeve prototype incorporated 16 vibrotactors and a control unit, providing precise, task-relevant haptic feedback specifically to space crew members in operating robotic arms. Vibrotactors were strategically spaced and positioned to cover the entire length of the forearm and the control unit is housed on the upper arm. Each vibrotactor was spaced approximately 3.0 cm apart horizontally and vertically, ensuring effective stimulus delivery while maintaining a low profile not to hinder movement (Figure 1a).

This arrangement supported both ATM and PS illusions to convey three-dimensional robotic movement and velocity information. We developed two rendering algorithms to translate end-effector movements in Euclidean space to the activation of vibrotactors on the forearm (Figure 1b). Movements in the positive and negative X-directions were encoded as sequential waves of vibration (ATM), traveling along the medial (positive X direction) or lateral portions (negative X direction) of the forearm. Movements in the Y-direction were represented by waves of vibration traveling up or down the dorsal side of the forearm through ATM with PS. For the Z-direction, movements were conveyed by ATM combined with PS illusions moving along both the medial and lateral portions of the forearm.

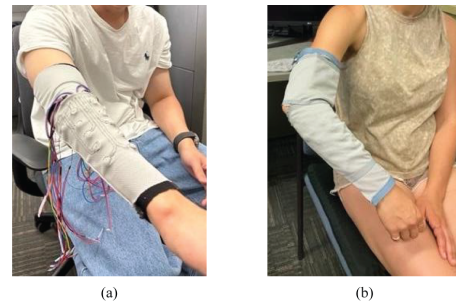


Figure 2: (a) First prototype of haptic sleeve, (b) Second prototype of haptic sleeve with e-textile lines on the forearm

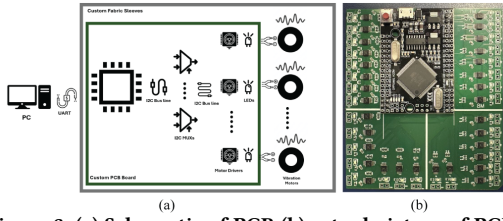


Figure 3: (a) Schematic of PCB (b) actual picture of PCB.

Positive Z-direction movements were indicated by vibration traveling from the ventral to the dorsal part of the forearm, while negative Z-direction movements were represented by vibration moving from the dorsal to the ventral part. The velocity of these tactile stroke illusions was manipulated at three levels by adjusting the duration time of individual vibrotactors and inter-stimulus onset asynchrony (SOA), which is the function of the duration time [21] (Figure 4).

To ensure wearability and comfort the haptic sleeve prototype went through 2 rounds of iteration. First, the sleeve was constructed from two layers of lightweight, breathable, stretchy knit fabric to maintain secure contact with the skin despite the lack of gravity. Vibrotactors and the cables connecting them to the control unit were securely sandwiched between the two layers of fabric in round pockets and channels (Figure 2a). In the second iteration, cables were eliminated and e-textile signal transmission lines were developed which allowed the use of a single layer of fabric (Figure 2b). In microgravity, the single layer and wire-free design ensured more consistent skin contact. The e-textile signal transmission lines were stitched on the sleeve with Karl Grimm copper Silver plated High Flex 3981 conductive thread and attached to the vibrotactors with round metal crimps and looped stitches. The sleeve also prioritized ease of movement and thermal comfort with an elbow opening (Figure 1c&d).

A compact control unit housed in a pocket on the upper arm managed the frequency, amplitude, and activation timing of each vibrotactor independently. The control unit ensured wireless communication with the robotic arm's control system to receive real-time operational commands, facilitating seamless integration. The sleeve was equipped with a rechargeable power source suitable for extended missions and low-latency wireless connectivity for smooth operation. Special attention was given to the placement and securement of the control unit to ensure it did not interfere with the astronaut's mobility or tasks. The haptic sleeve successfully provided real-time haptic feedback corresponding to the robotic arm's movements and interactions.

3.2 Design of Electronics Hardware

We developed a specialized electronic hardware platform (shown in Figure 3) centered on the ATmega2560 microcontroller (Microchip Technology Inc.) for managing precise vibrotactor actuation, where the microcontroller communicated directly with a computer via a UART protocol. To efficiently handle multiple control inputs and outputs, our system incorporated four TCA9548APWR I2C multiplexer chips (Texas Instruments Incorporated). These chips facilitate the distribution of I2C data from the microcontroller to 32 possible vibrotactor motor drivers, although we employed 16 vibrotactors

in this study. The DRV2605LDGSR motor drivers were chosen for their ability to interpret a range of control signals such as voltage, frequency, and duration (Texas Instruments Incorporated), ensuring precise motor operation. An LED was integrated parallel to each motor's output, allowing operators to visually confirm motor functionality directly.

4 PILOT USER TESTING

4.1 Design of the feasibility test

To examine the feasibility of using the haptic sleeve to encode robotic movement information, we recruited N=10 participants to perform testing. Our protocol was approved by the UC Davis Institutional Review Board (IRB), and participants provided written informed consent.

Prior to testing we measured the maximum circumference of each participant's self-reported dominant arm's forearm. Testing then began with participants donning the haptic sleeve on their self-reported dominant arm. They were informed that the testing would involve patterns of vibration displayed to their forearm via the haptic sleeve. This vibration would encode directional information, which included orthogonal directions ($\pm X$, Y , or Z) at three possible velocities (slow, medium, or fast, as described in the design section above), totaling 18 combinations of direction and velocity. Participants were informed that we would provide a vibration pattern to their arm, and their task was to report the direction and velocity at which they perceived the encoded movement. They were first provided with a familiarization period during which each combination of direction and velocity was sequentially displayed to the participant via the haptic sleeve, and they were informed of the direction and velocity each combination encoded. Testing then began with each combination of direction and velocity being randomized and repeated 3 times, resulting in a total of 54 vibration patterns presented to each participant.

Our primary outcome measure was the recognition accuracy with which participants could report the direction and velocity of vibration patterns. To identify significant differences in our data, we employed a multiple linear mixed-effects (LME) model. In this model, the accuracy of participants' responses was the dependent variable, with combinations of direction and speed serving as fixed effects and individual participants included as a random effect.

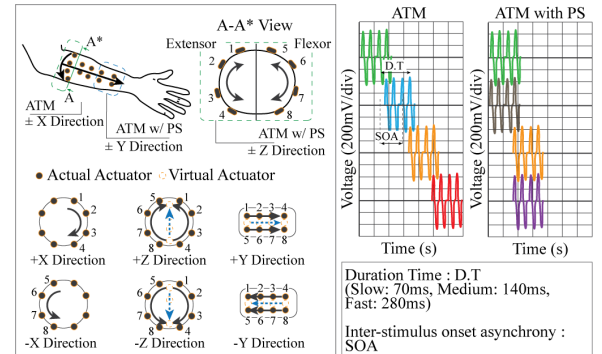


Figure 4: Schematic description for conveying directional and speed information through the sequential activation of vibrotactors arranged in a haptic sleeve.

Furthermore, we analyzed the comparison of individual fixed effects and their interactions using false discovery rate adjustment after obtaining estimated marginal means.

4.2 Findings and Discussion

Figure 5 shows the recognition accuracy for each type of information individually (direction and velocity). Participants correctly reported the directional patterns with approximately 90% accuracy. Participants correctly identified the patterns in the negative X direction and the positive and negative Y directions with a median accuracy value of 100%. However, there was greater variability in the positive X direction and the positive Z direction, with a median accuracy value of 94.44% (Q1=88.89%, Q3=100%), and in the negative Z direction, with a median accuracy value of 100% (Q1=88.89%, Q3=100%). However, these differences were not statistically significant. These results may reflect the variability of vibration perception across various dermatomes on the forearm (lateral (C7) and medial (T1) portions of the forearm). This could be attributed to the difference in the extent of somatosensory cortical representation in the human brain or known differences in the density of mechanoreceptors in these forearm areas [9]. Additionally, participants reported higher accuracy for the fast speed pattern compared to other speeds (medium and slow), with a statistically significant difference between fast and medium ($p < 0.0001$) and between fast and slow speed ($p < 0.0001$).

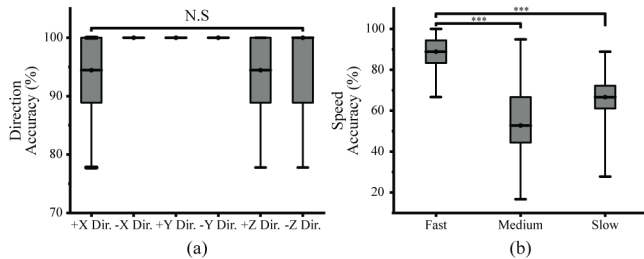


Figure 5: Recognition accuracy for each vibration pattern (a) direction and (b) speed, the median value indicated on the box as a black circle, and the whiskers extended from the minimum to the maximum. (* $p < 0.5$, ** $p < 0.01$, * $p < 0.001$).**

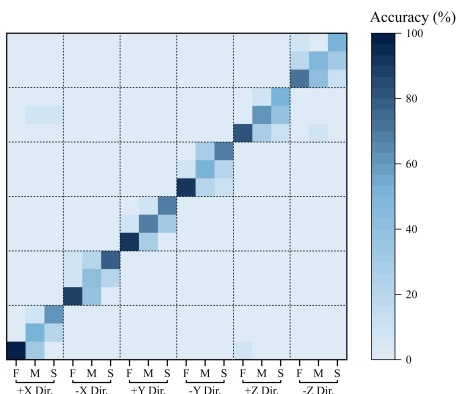


Figure 6: Confusion matrix of recognition accuracy for each vibration pattern combined with directions and speed

We created a confusion matrix to illustrate participants' recognition average accuracy across all 18 direction and velocity combinations. Here it is possible to see that participants were readily

able to identify Euclidian directional information and that the fast velocity encoding outperformed the medium and slow (Figure 6).

Finally, to examine if the fit of the haptic sleeve influenced accuracy results, we performed a correlation analysis. Participants' forearm circumferences ranged from 25.4 cm to 30.2 cm (mean = $27.4 \text{ cm} \pm 2.0 \text{ cm}$). We found a very weak negative correlation ($r = -0.036$) between the combined direction-and-velocity results and forearm circumference. However, it should be noted that the goal of the human tests was to examine the feasibility of the haptic sleeve, and the range of forearm sizes tested does not necessarily represent the diversity anticipated across flight crew members. Thus, further testing to examine the effects of fit is warranted, as is the examination of sleeve sizing and adjustable features to accommodate varying forearm sizes.

5 CONCLUSION AND FUTURE DIRECTIONS

In this paper, we contributed to the development of a prototype wearable haptic sleeve designed to induce tactile stroke illusions and feasibility testing with human subjects. Our feasibility test demonstrated its potential for conveying task-relevant information to robotic operation movements. Specifically, we arrived at two key findings: first, participants demonstrated an acute ability to use tactile stroke illusions to interpret Euclidean directional cues; second, participants were readily able to detect differences in velocity information; accurately identifying fast velocity separate from slow or medium. However, we noted that the medium and slow velocity patterns were less accurately perceived which opens new directions for future exploration. For example, future experiments may focus on how best to manipulate vibration profiles or the resolution to which we can detect velocity differences. This may help research identify strategies to maximize the amount of information that can be encoded through tactile stroke illusions. Collectively these results open new opportunities for haptic augmentation during robotic operations, aiming to improve the efficacy of performance in microgravity (or non-terrestrial) operations and potentially reduce reliance on highly taxed visual sensory channels.

In addition, future research will focus on refining the prototype to enhance its reliability and user experience. We plan to conduct extensive testing by manipulating various physical characteristics of vibration to evaluate its effectiveness in conveying speed and complex robotic movements. To achieve these future directions, we will explore vibration perception across parts of the forearm to induce tactile stroke illusions and validate the methods for rendering tactile sensations using this sleeve with a much larger subject population. Moreover, since we now have this sleeve platform, we will also explore the simultaneous collection[12, 13, 20, 25] and analysis[6, 13, 18] of physiological signals during feedback. Additionally, we will assess the efficacy of haptic feedback in learning to control robotic operations during space-mission-relevant training activities. Ultimately, our goal is to develop a robust and intuitive haptic interface that significantly enhances the precision and efficiency of robotic operations in non-earth environments.

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